

Article

Analysis of Industrial Diversification Level of Economic Development in Rural Areas Using Herfindahl Index and Two-Step Clustering

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Abstract: The purpose of this study is to analyze the possible relationship between industrial structure and economic development in rural areas in South Korea. Accordingly, this study uses the Herfindahl–Hirschman Index and a two-step cluster analysis method to conduct an empirical analysis of the rural areas of Chungcheongbuk-do as the research object. The results show that among the 11 regions with concentrated industrial structures, the cluster results of 2 regions changed from the decentralized low employment cluster in 2010 to a concentrated high employment cluster in 2015, while the cluster results of other regions remained unchanged. Among the 18 regions with decentralized industrial structure, the cluster results of 5 regions changed from the concentrated high employment cluster in 2010 to the decentralized low employment cluster. Meanwhile, the cluster results of three regions changed from the decentralized low employment cluster in 2010 to the concentrated high employment cluster in 2015, while the cluster results of other regions remained unchanged. Based on this, it can be concluded that, for general rural areas, a low level of industrial diversification, that is, a concentrated industrial structure, is more conducive to promoting the economic development of rural areas. However, there is a special case, namely that rural areas with certain specific advantages, a high level of industrial diversification, or a decentralized industrial structure are more conducive to the development of the regional economy.

Keywords: rural areas; industrial structure; economic development; HHI index; two-step cluster analysis



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1. Introduction

Since 2000, South Korea has been transforming into an aging society, with the situation becoming grave in rural areas [1]. Rural areas are facing economic stagnation driven by a shortage of labor force caused by an aging population [2]. Gao et al. (2018) studied the main reasons for China's rural labor outflow. Factors, such as the urban–rural income gap, poverty, reduced labor demand in rural areas, underdeveloped infrastructure, the low quality of social services in rural settlements, and the low social status of rural residents have contributed to the rural labor outflow. It is also believed that the fundamentals of sustainable rural development rely on the retention of the rural labor force, and the diversification of the rural economy helps to retain the labor force in the rural settlements [3]. Gheorghe Chiru (2021), in his article on the development of entrepreneurship in rural areas of Romania, believes that the improvement of the performance of entrepreneurship in rural areas is the main means to narrow the existing gap between urban and rural areas, and local entrepreneurship in rural areas is considered to be one of the key factors in achieving sustainable development in rural areas [4]. Lu Ying et al. (2022) used quantitative methods

to analyze the discourse transformation in China's rural development in the past 40 years, distinguished eight rural development paradigms, believed that the key feature of the current rural development discourse is duality, and emphasized the importance of balanced development. They also believe that township enterprises have played an important role in rural economic development [5]. Lewandowska et al. (2021) believe that policies, plans, and actions focusing only on entrepreneurship and innovation are doomed to fail if they do not consider the particularity of regions. Therefore, policies to effectively promote the development of SMEs should also include empirical methods and consider regional specificity [6]. Industrial agglomeration and its externalities support the development of regional economies [7]. The premise of promoting economic development in rural areas is to appropriately interpret the current situation of industrial agglomeration in these areas and to clarify the relationship between industrial agglomeration and regional economic development. Yuan points out that the externalities of industrial agglomeration are composed of Marshall's specialized industrial agglomeration and the diversified industrial agglomeration of Jacobs [8]. These two externalities generate knowledge spillovers through competition and integration among enterprises, and accelerate innovation to promote regional economic development. Marshall (1890) first proposed that the agglomeration economy originates from externalities generated by the clustering of enterprises in similar industries. He believed that the agglomeration of enterprises in similar industries not only helps enterprises to share professional labor and intermediate input factors to reduce production costs, but also that the competitive effect caused by the agglomeration of enterprises in the same industry should accelerate knowledge spillover and technological innovation to improve labor productivity and promote regional economic development. Jacobs (1969) believes that the externalities of an agglomeration economy benefit from the amalgamation of enterprises in different industries, and the complementarity and diversity of multiple industries are conducive to promoting knowledge integration and the collision of ideas, emphasizing the role of multi-industry agglomeration on regional innovation and regional economic development. South Korea has produced vast research on the relationship between industrial agglomeration and economic development in urban areas, but there are limited studies on the current situation of industrial agglomeration and its relationship with regional economic development in rural areas. Shim (2008), Moon et al. (2015), Kim et al. (2018), and Zheng et al. (2021) have reached similar conclusions on the impact of industrial diversification on urban economic development. They all believe that diversified industrial structure contributes to the needs of economic development of large cities, while, for small cities, specialized industrial structure can stimulate economic vitality [1,7,9,10]. The purpose of this article is to explore the potential relationship between diversified industrial agglomeration and regional economic development in rural areas, that is, whether a high or low level of industrial diversification is beneficial to economic development. Compared with large cities, the economic development of rural areas should be closer to that of small cities. Therefore, this paper proposes the following hypothesis: For rural areas, the industrial structure with a low level of diversification is more conducive to the development of regional economy. Therefore, this study takes the rural areas of Chungbuk-do as the research object and obtains relevant data, such as industrial sales and employees in rural areas of 2010 and 2015, according to the general economic survey provided by the National Bureau of Statistics of Korea. First, based on the industrial sales data of each rural area, the HHI value of each area was calculated using the Herfindahl–Hirschman Index. The HHI is an index that can describe diversified industrial agglomeration [11]. The closer the HHI value is to 1, the lower the level of regional industrial diversification; that is, the more concentrated the regional industrial structure, while the closer the HHI value is to 0, the higher the level of regional industrial diversification, that is, the more decentralized the industrial structure. The ArcGIS software was employed to draw the HHI distribution map of the rural areas of Chungbuk in 2010 and 2015, and to analyze the current situation and the changing trends of the regional industrial structure. Subsequently, the HHI value and the number of employees in each

region were used as continuous variables, and the top three major industries in each region as categorical variables to conduct a two-step cluster analysis, through the analysis of the changes in cluster results in areas where industrial structure has changed, to explore the relationship between industrial structure and economic development in rural areas. Accordingly, we propose an industrial strategy suitable for economic development in rural areas. This article has certain reference significance for the Korean and local governments to introduce related industrial policies to enhance rural economies.

2. Materials and Methods

2.1. Research on Industrial Agglomeration and Its Relationship with Economic Development

Industrial agglomeration is the macro-reaction of the production and consumption behaviors of enterprises, producers, and consumers in a spatial agglomeration. The main reason for spatial agglomeration is to obtain a positive externality. The spatial externality of industrial agglomeration is affected by the agglomeration model, which is categorized into specialized and diversified industrial agglomerations. Specialized industrial agglomeration refers to the degree of agglomeration of a certain advantageous industry in a specific space, while diversified industrial agglomeration reflects the level of industrial diversification in a specific space [12].

Specialized industrial agglomeration was first proposed by Marshall (1920), who believes that the agglomeration economy originates from externalities generated by the clustering of companies in the same industry. He believes that the agglomeration of companies in the same industry not only helps companies to share professional labor and intermediate input factors to reduce production costs, but also that the accompanying competitive effect will accelerate knowledge spillover and technological innovation, thus improving labor productivity and promoting regional economic development.

Diversified industrial agglomeration was first proposed by Jacobs (1969). He believes that the externalities of an agglomeration economy benefit from the coexistence of companies in different industries. The complementarity and diversity of multiple industries can promote knowledge integration and innovation, emphasizing the role of multi-industry agglomeration in local innovation and economic development. In contrast to the clear description of specialized agglomeration, the proposal of diversified agglomeration has experienced a complicated development path. Finally, Parr clearly describes the agglomeration of diversified industries with industry types and industry distribution balances. He believes that the more types of industries there are in the same area, the more balanced the distribution of industries is, indicating a higher degree of industrial diversification in the region.

Based on the above theory of industrial agglomeration, the research results of scholars are summarized as follows. Ryu & Yoon believed that for a wide area, industrial structure diversity has a positive impact on economic growth [13]. Moon et al. (2014) explored the relationship between the diversification of industrial structure and regional economic growth based on the diversified industrial agglomeration theory proposed by Jacobs [11]. The study concludes that for autonomous organizations in the Gyeonggi region, rather than focusing on the development of certain industries, increasing industrial diversification can increase productivity and promote regional economic development. Moon et al. (2015) analyzed the impact of industrial diversification on the economic growth of cities of different sizes and locations [10]. The results of the study indicated that for the capital area, the improvement in the level of industrial diversification can promote the development of the regional economy; for non-capital areas with a population of more than 300,000, industrial specialization contributes to the development of the regional economy; for cities with a population of less than 300,000, the diversification of industries has no significant impact on regional economic growth. Kim et al. (2018) studied the impact of industrial concentration and diversification on the economic growth of local cities and concluded that in times of economic recession, specialized industrial development strategies have a greater negative externality on the regional economy [9]. Zheng et al. (2021) conducted an

empirical analysis on the relationship between industrial agglomeration, urban population size, and economic development, and concluded that cities that are midsized and above should choose a diversified industrial agglomeration model, and that the “small and fine” specialized industrial agglomeration model is more consistent with the economic development of small cities [7]. Shim (2008) conducted a comparative analysis of the changes in industrial structure in rural and urban areas based on the changes in population structure, such as low birth rates and aging. The results of the analysis show that as the low birth rate and degree of aging intensify, specialized industries will gradually collapse in Chungcheongnam-do (rural) areas, while those in the Daejeon (urban) area will increase; that is, the aggravation of low birth rate and aging has a significant impact on the industrial structure between urban and rural areas [1].

By combining the existing findings, it can be determined that research on industrial agglomeration mainly focuses on the urban level, with limited focus on rural areas. Therefore, this study focuses on rural areas and attempts to explore the possible relationship between industrial diversification and economic development in rural areas based on Jacobs’s theory of diversified industrial agglomeration. In this study, a high level of industrial diversification is defined as a decentralized industrial structure, and a low level is defined as a concentrated industrial structure.

2.2. Herfindal-Hirschman Index (HHI)

Industrial agglomeration is measured by HHI, which refers to the sum of squares of the market shares of enterprises in a particular market. Rhoades used HHI to analyze the market share of specific industries [14]. Gambardella & Torrisi (1998) used HHI to judge the degree of technological integration in the electronic information industry, while Lu et al. (2017) evaluated the performance of convergence development based on HHI, and Susilo & Axhausen (2014) used HHI to analyze personal activity, travel, and location patterns [15–17]. The formula is as follows:

$$HHI = \sum_{i=1}^n \left(\frac{X_i}{X} \right)^2$$

At present, HHI is a widely used index to measure the agglomeration of diversified industries. The HHI values range between 0 and 1. The closer the HHI value is to 1, the lower the level of industrial diversification, and the closer the HHI value is to 0, the higher the level of industrial diversification. Therefore, the HHI index can be used to measure the level of industrial diversification. Moon et al. (2014) employed Jacobs’s theory of diversified industrial agglomeration to explore the relationship between the diversification of industrial structure and regional economic growth by using the HHI of the number of employees to measure industrial diversification. In his research, the 1-HHI value is used to indicate the level of industrial diversification to facilitate understanding; that is, the greater the 1-HHI value, the higher the level of industrial diversification [11]. Zheng (2021) conducted an empirical analysis of the relationship between industrial agglomeration, urban population size, and economic development, using the reciprocal of HHI, i.e., $1/HHI$, to measure the degree of industrial diversification agglomeration [7]. Kim et al. (2018) studied the impact of industrial concentration and diversification on the economic growth of local industrial cities, and then used HHI to measure the level of industrial diversification. The smaller the HHI value, the higher the level of industrial diversification [9].

In this study, the value range of HHI was divided into five sections, as follows: 0.81–1.00, 0.61–0.80, 0.41–0.60, 0.21–0.40, and 0.01–0.20. Section 0.81–1.00 represents a high concentration, 0.61–0.80 a medium high concentration, 0.41–0.60 a medium concentration, a 0.21–0.40 medium high decentralization, and 0.01–0.20 a high decentralization. The specific details are listed in Table 1. Based on industrial sales data of rural areas, this study measures the industrial structure of rural areas using HHI.

Table 1. HHI value division interval.

HHI Interval	0.01–0.20	0.21–0.40	0.41–0.60	0.61–0.80	0.81–1.00
Concentration			Medium concentration	Medium high concentration	High concentration
Decentralization	High decentralization	Medium high decentralization			

2.3. Two-Step Cluster Analysis

The two-step cluster analysis method helps gather similar objects to form a cluster. Compared with other clustering methods, the advantage of the two-step clustering analysis method is that it can cluster categorical and continuous variables simultaneously and can deal with large-volume sample data. Kim & Hwang (2017) used a two-step clustering method to analyze the characteristics of rural tourists to segment the tourism market [18]. When analyzing the status quo of industries in rural areas, it is not only necessary to analyze the industrial structure of the region but also to identify which are the main industries. Therefore, the two-step clustering method was selected for the cluster analysis of industrial structures in rural areas. Although rural areas are involved in various industries, this study focuses on the top three industries with a high proportion, that is, the categorical variables are the top three industries, while the continuous variables include HHI values and the number of employees. The number of employees is an indicator of the economic situation. To a certain extent, more employees can explain better regional economic development. This paper presents sample data from 102 rural areas in Chungcheongbuk-do in SPSS for a two-step cluster analysis. The number of clusters was automatically determined using the Bayesian inference criterion (BIC) value as the clustering criterion. Noise processing was implemented to eliminate outliers, and the log-likelihood distance was used to calculate the distance. Finally, according to the BIC, the analysis results were divided into the following two clusters: concentrated high employment clusters and decentralized low employment clusters. Thereafter, we compared the clustering results of the areas where the industrial structure has changed to indirectly reflect the possible relationship between the industrial structure and economic development in rural areas.

The research path of this article is shown in Figure 1:

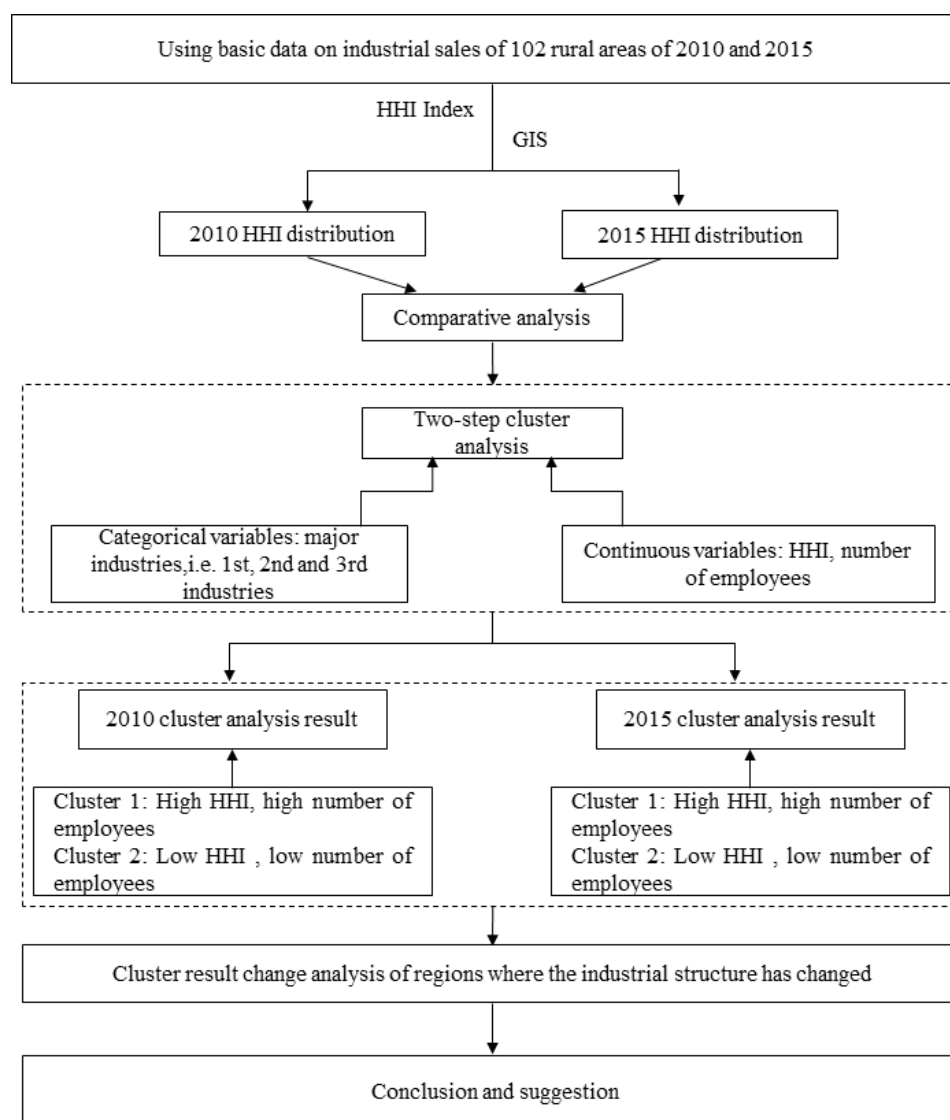


Figure 1. Research roadmap for two-step clustering using HHI.

3. Results and Discussion

3.1. Study Area and Data

This study explores the possible relationship between industrial structure and economic development in rural areas. Therefore, this study considers the rural areas of Chungcheongbuk-do, that is, 102 rural areas of eup and myeon belonging to three cities and eight counties in Chungcheongbuk-do, as the research object for empirical analysis. A general economic survey covering the areas of eup and myeon has been implemented every five years since 2010. Therefore, based on the general economic survey data of 2010 and 2015 provided by the Korean Bureau of Statistics, this study extracted the industrial sales and number of employees in various rural areas of Chungbuk. In the general economic survey, the industry classification standard is the Korean standard industrial classification, which contains 21 major classifications, except for self-consumption (T) and foreign institutions (U). The names of the industry classifications in 2010 and 2015 are marginally different. In this study, the name of the industry classification of the 2015 general economic survey is used as the standard, and the specific classification is shown in Table 2.

Table 2. Korean standard industry classification.

Code	Industrial Classification	Code	Industrial Classification
A	Agriculture, forestry, and fisheries	K	Finance and insurance industry
B	Mining	L	Estate
C	Manufacturing industry	M	Professional and technology services
D	Electrical, gas, steam, and air conditioning supply industry	N	Business facility management and business support and leasing services
E	Tap water, sewage and waste treatment, raw material recycling industry	O	Public administration, national defense, and social security administration
F	Construction industry	P	Education service industry
G	Wholesale and retail	Q	Health care and social welfare services
H	Transportation and storage industry	R	Arts, sports, and leisure related services
I	Accommodation and catering industry	S	Associations and clusters, repair, and other personal services
J	Information and communication industry		

To correctly analyze the industrial structure of rural areas and its possible relationship with economic development, this study considers the rural areas of Chungcheongbuk-do as the research object. Finally, the relevant data, such as industrial sales and employees in Chungbuk rural areas of 2010 and 2015, were obtained according to the general economic survey provided by the Korean Bureau of Statistics. Based on the industrial sales data of rural areas, the industrial structure of rural areas is measured by HHI, and the change trend of the industrial structure is analyzed. To analyze the relationship between regional industrial structure and economic development, this study indirectly reflects the possible relationship between the two by analyzing the changes in the cluster results of areas where the industrial structure has changed.

3.2. Analysis of the Industrial Structure and Its Changing Trend in Rural Areas

Based on industrial sales data from various rural areas, the HHI values of each region were calculated, which ranged from 0 to 1. The closer the HHI value is to 1, the more concentrated the regional industrial structure, and the closer the HHI value is to 0, the more decentralized the regional industrial structure. Based on the calculated HHI value, the distribution map of HHI in the Chungbuk rural area in 2010 and 2015 was drawn using ArcGIS, as shown in Figures 2 and 3 below, and the HHI value, sales, and the number of employees were analyzed using descriptive statistics, as shown in Tables 3 and 4.

Table 3. Data of 2010 descriptive statistical analysis.

	Sales (Million Won)	HHI	Number of Employees
Mean	577,806	0.46	2336
Std. Deviation	1,048,131	0.24	3115
Median	138,676	0.40	961
Minimum	4194	0.11	93
Maximum	6,351,427	0.96	17,070
N	102	102	102

Table 4. Data of 2015 descriptive statistical analysis.

	Sales (Million Won)	HHI	Number of Employees
Mean	863,155	0.43	3138
Std. Deviation	1,509,069	0.23	4401
Median	247,451	0.38	1311
Minimum	6715	0.12	122
Maximum	7,604,424	0.97	26,006
N	102	102	102

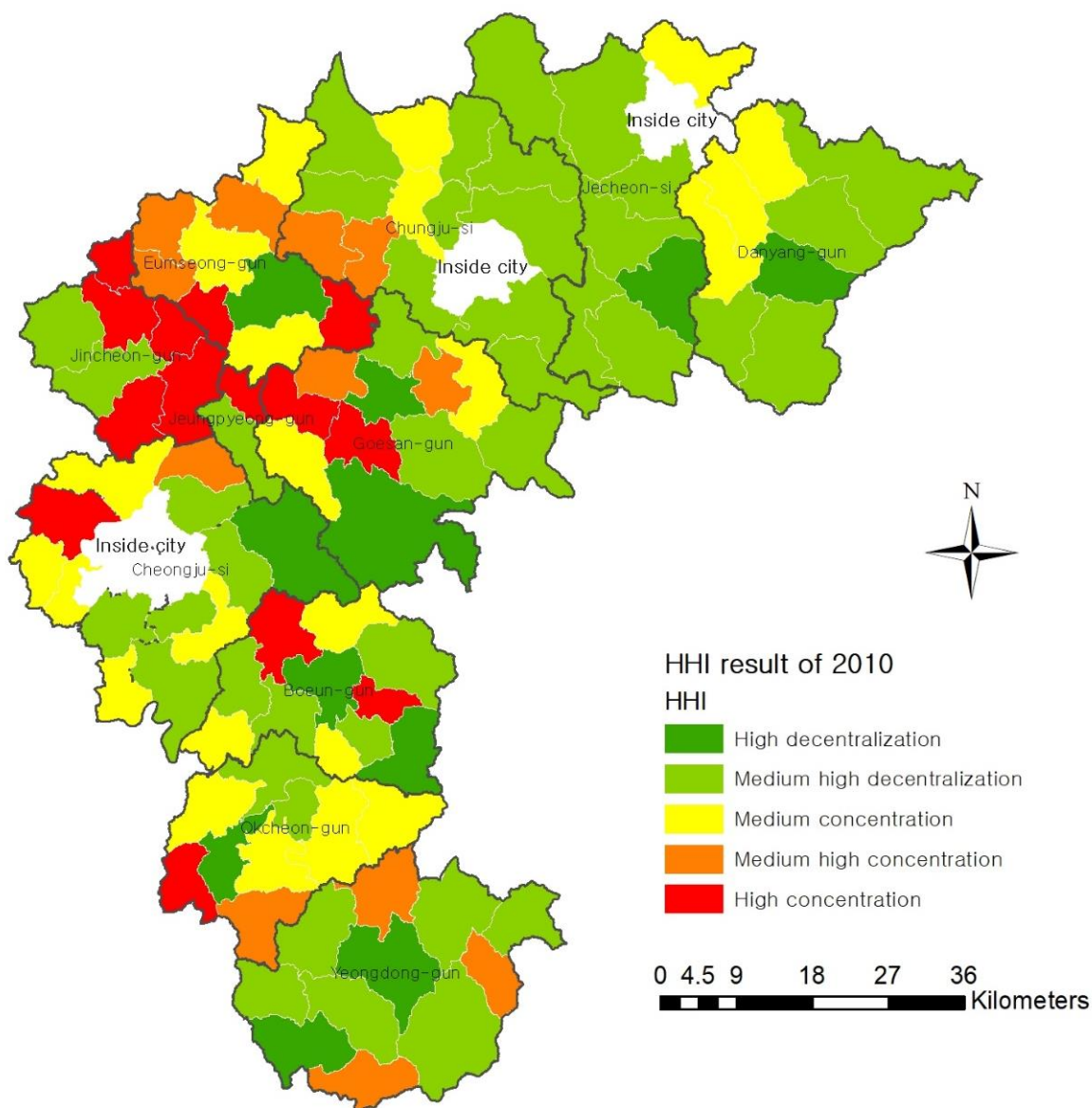


Figure 2. The HHI distribution map of 2010.

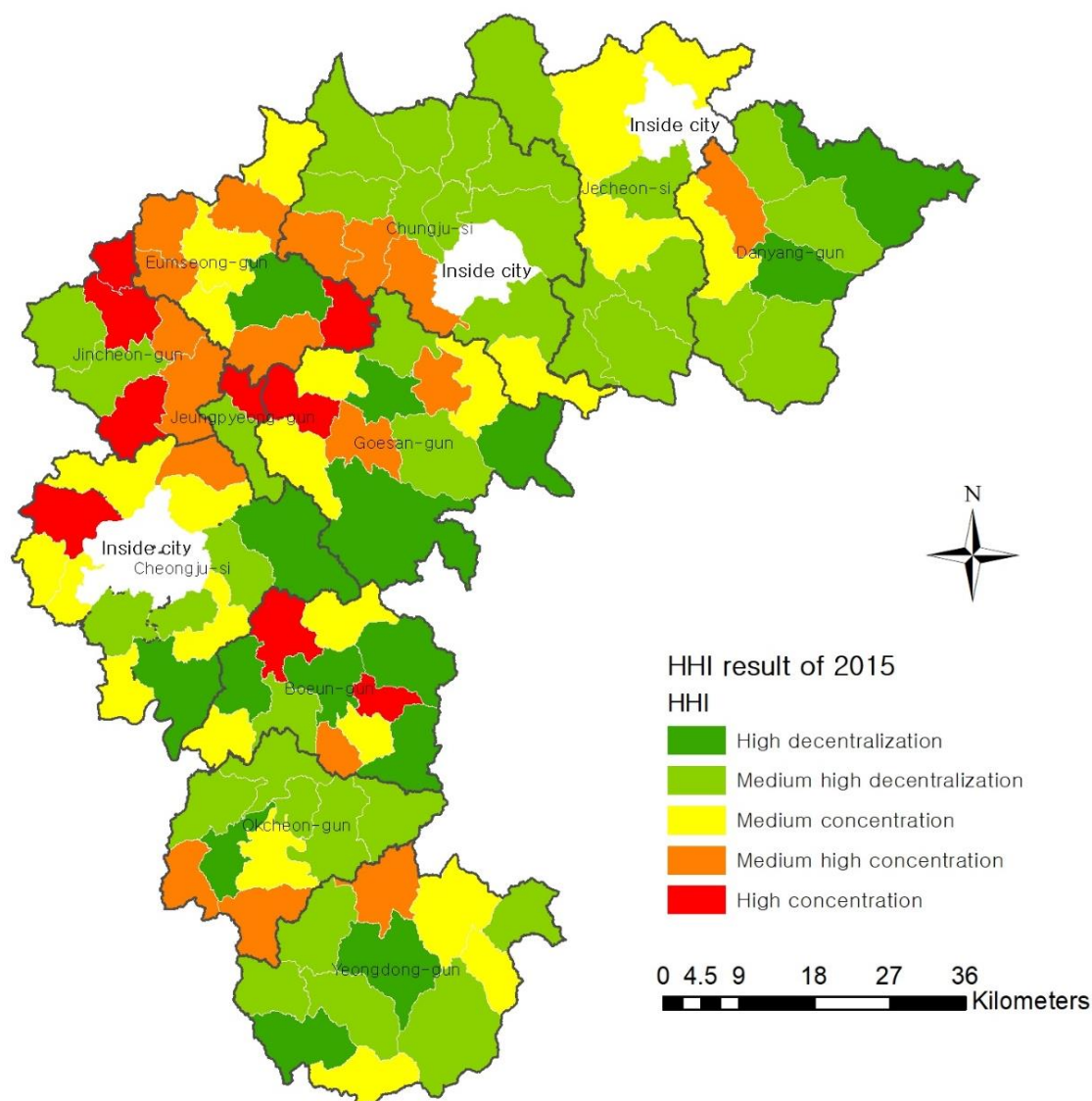


Figure 3. The HHI distribution map of 2015.

3.2.1. Descriptive Statistical Analysis

As shown in Tables 3 and 4, it can be found that there is little difference in the mean, standard deviation, maximum, minimum, and median of HHI in 2010 and 2015, indicating that, compared with 2010, the industrial structure of most rural areas in Chungbuk in 2015 did not change. In contrast, in terms of the number of employees, the average number of employees in 2015 was 3138, up from 2336 in 2010. The standard deviations of both are large, indicating that the gap in the number of employees in various regions is wide, and that the level of economic development among regions is uneven. Compared to 2010, although the industrial structure of most rural areas in Chungbuk did not change significantly in 2015, the industrial structure of some areas changed considerably.

3.2.2. Comparative Analysis of Industrial Structure in the Rural Areas of Chungbuk

The HHI distribution maps for 2010 and 2015 are shown in Figures 2 and 3. The comparison shows that the industrial structure of most regions has not changed, but that the industrial structure of some regions has undergone significant changes. The specific contents are as follows.

Among the rural areas in Danyang-gun, the industrial structure of the Maepo-eup area shows a trend of concentration, while the industrial structure of Eosangcheon-myeon and

Yeongchun-myeon shows a trend of decentralization, and the industrial structure of other areas shows no obvious changes. The reason for the concentration of the industrial structure of the Maepo-eup area is the rise in the proportion of the manufacturing industry from 70.23% in 2010 to 79.98% in 2015. The proportion of the construction industry dropped sharply, moving down from the top three industries. The industrial structure of the Eosangcheon-myeon area shows a trend of decentralization, because the proportion of manufacturing slumped from 59.47% in 2010 to 27.47% in 2015, and the mining industry rose from 23.65% in 2010 to 35.74% in 2015. Meanwhile, the proportion of the public administration industry increased to 13%. The reason for the decentralization of the industrial structure in the Yeongchun-myeon area is that the proportion of the financial and insurance industries has dropped sharply, moving down from the top three. In 2015, the proportion of the wholesale and retail industry was 37.64%, while the share of the education service industry increased to 13.34%, and that of the accommodation and catering industries rose to 10.58%.

In the rural areas of Jecheon-si, the industrial structures of Bongyang-eup, Cheongpung-myeon, and Susan-myeon show a trend of centralization, while the industrial structures of other regions show no obvious changes. The industrial structure of the Bongyang-eup area shows a trend of concentration because the proportion of manufacturing has increased from 57.32% in 2010 to 63.68% in 2015; that is, the region is concentrated in manufacturing. The industrial structure of the Cheongpung-myeon area shows a trend of concentration because the proportion of the catering and accommodation industry has risen sharply from 49.37% in 2010 to 68.86% in 2015; that is, the region is concentrated in the catering and accommodation industries. The industrial structure of the Susan-myeon area shows a trend of concentration because the proportion of wholesale and retail industries increased from 27.01% in 2010 to 33.77%, and that of catering and accommodation industries rose from 18.23% in 2010 to 26.87% in 2015; that is, the region concentrates on the development of wholesale and retail, catering, and accommodation industries. These areas focus on the development of major regional industries.

In the rural areas of Chungju-si, the industrial structure of Sotae-myeon and Jungangtap-myeon shows a trend of decentralization, while the industrial structure of Daesowon-myeon and Suanbo-myeon shows a trend of centralization, and the industrial structure of other regions shows no obvious trend of change. The industrial structure of the Sotae-myeon region shows a decentralization trend because the proportion of the manufacturing industry has declined from 65.6% in 2010 to 47.03% in 2015. In 2015, agriculture, forestry, and fisheries accounted for 31.45% of the region's newly emerging industries. The industrial structure of the Suanbo-myeon region shows a trend of centralization because the proportion of the education service industry has risen sharply from 37.85% to 72.15%. The industrial structure of the Daesowon-myeon region shows a trend of concentration because the share of manufacturing has increased substantially. In 2010, the manufacturing industry accounted for 51.33%, and in 2015, it accounted for 79.01%. The proportion of wholesale, retail, and educational services have both dropped sharply, indicating that the industry in the region is concentrated in manufacturing.

Among the various rural areas in Eumseong-gun, the industrial structure of the Maengdong-myeon area shows a decentralization trend, while that of Wonnam-myeon shows a concentration trend, and the industrial structure of other areas shows no obvious change. The reason for the decentralization of the industrial structure in the Maengdong-myeon region is the sharp decline in the proportion of manufacturing, which dropped from 91.34% in 2010 to 65.6% in 2015, leading to an increase of 10.82% in the public administration industry. The industrial structure of the Wonnam-myeon region shows a trend of centralization because the proportion of manufacturing has risen sharply, from 62.64% in 2010 to 85.23% in 2015, and that of wholesale and retail industries has dropped from 16.83% to 4.47%; that is, the region concentrates on manufacturing development.

In the rural areas of Jincheon-gun, the industrial structures of Deoksan-myeon and Chopyeong-myeon show a trend of decentralization, whereas those of other regions show

no obvious change. The reason for the decentralization of the industrial structure in the Deoksan-myeon region is that the proportion of manufacturing declined from 96.32% in 2010 to 86.83% in 2015, while the proportion of wholesale and retail industries increased from 1.68% in 2010 to 8.33%. The reason for the decentralization of the industrial structure in the Chopyeong-myeon region is that the share of manufacturing declined from 90.73% in 2010 to 82% in 2015, while that of wholesale and retail increased from 4.41% in 2010 to 9.69% in 2015. The main reason for the decentralization of the industrial structure in these regions is the reduction in the scale of manufacturing, and the expansion of the scale of the service industry.

In the rural areas of Goesan-gun, the industrial structures of Yeonpung-myeon, Mungwang-myeon, and Sosu-myeon show a trend of decentralization, and the industrial structure of other regions shows no obvious change. The industrial structure of the Yeonpung-myeon region shows a decentralized trend because the main industry in the region was manufacturing in 2010, accounting for 34.93%, whereas in 2015, the main industry was finance and insurance, accounting for 26.81%, while agriculture, forestry, and fisheries accounted for 22.55%, and the wholesale and retail industry 18.41%. The proportion of each industry is relatively balanced. The reason for the decentralization of the industrial structure in the Mungwang-myeon region is that the share of manufacturing declined from 90.1% in 2010 to 80.07% in 2015, while that of wholesale and retail increased to 10.81%. The reason for the decentralization of the industrial structure of the Sosu-myeon region is that the proportion of manufacturing declined from 84.46% in 2010 to 70.89% in 2015, while the proportion of wholesale and retail increased to 28.45%. The main reason for the decentralization of the industrial structure in these regions is the reduction in the scale of manufacturing and the expansion of the scale of the service industry.

In the rural areas of Cheongju-si, the industrial structure of Munui-myeon shows a trend of decentralization, while the industrial structure of Naesu-eup demonstrates a concentration trend, and other regions show no obvious change. The industrial structure of the Munui-myeon region shows a decentralized trend because the share of manufacturing declined from 49.36% in 2010 to 31.27% in 2015, and the share of education services increased to 14.25%. The industrial structure of the Naesu-eup area shows a trend of concentration because the proportion of manufacturing has increased from 55.05% in 2010 to 61.53% in 2015; that is, the region's industries are developing a focus on manufacturing.

In the rural areas of Boeun-gun, the industrial structures of Samseung-myeon and Tanbu-myeon show a trend of concentration, while the industrial structure of Songnisan-myeon and Hoein-myeon shows a trend of decentralization, and the industrial structure of other regions shows no obvious change. The industrial structure of the Samseung-myeon region shows a trend of centralization, with the share of manufacturing increasing from 73.27% in 2010 to 78.01% in 2015, and the share of educational services rising to 6.14%. The industrial structure of the Tanbun-myeon region shows a trend of centralization, with the proportion of manufacturing increasing from 43.75% in 2010 to 60.28% in 2015, and that of agriculture, forestry, and fisheries declining from 27.08% to 18.44% during the period. Specifically, the industry in this region is concentrated in manufacturing. The industrial structure of the Songnisan-myeon area shows a weak decentralization trend, and the proportion of the leisure service industry increased to 12.56% in 2015, making it a potential industry. The industrial structure of the Hoein-myeon region shows a decentralization trend, considering that the share of the manufacturing industry has declined significantly, moving down from the top three, while the share of the healthcare industry increased to 27.77%, ranking first.

In the rural areas of Okcheon-gun, the industrial structures of Gunseo-myeon, Gunbuk-myeon, Cheongseong-myeon, and Cheongsan-myeon show a trend of decentralization, whereas those of other regions show no obvious change. The reason for the decentralization of the regional industrial structure in the Gunseo-myeon region is the sharp decline in the share of manufacturing from 96.61% in 2010 to 88% in 2015. The reason for the decentralization of the regional industrial structure in the Gunbuk-myeon region is that the

share of manufacturing dropped sharply, from 60.57% in 2010 to 47.67% in 2015, while the share of wholesale and retail increased to 25.54%. The decentralization of the industrial structure in the Cheongseong-myeon region can be attributed to the sharp decline in the proportion of the manufacturing industry from 64.45% in 2010 to 48.62%. In 2015, the transportation and storage industry in the region increased to 17.39%, becoming a newly emerging industry. The reason for the decentralization of the industrial structure in the Cheongsan-myeon region is the sharp decline in the proportion of manufacturing from 70% in 2010 to 56.62% in 2015, and an increase in the share of other industries. The main reason for the decentralization of the industrial structure in these regions is the reduction in the scale of manufacturing and the expansion of the scale of the service industry.

Among the rural areas in Yeongdong-gun, the industrial structure of Hwanggan-myeon shows a trend of concentration, while that of Maegok-myeon and Yonghwa-myeon shows a trend of decentralization, and the industrial structure of other regions shows no obvious change. The industrial structure of Hwanggan-myeon shows a centralization trend because the proportion of manufacturing has risen sharply, from 51.03% in 2010 to 62.65% in 2015. The proportion of science and technology services dropped sharply, moving down from the top three, and the wholesale and retail industries increased to 19.2%, that is, the region focused on the development of manufacturing, and the wholesale and retail industries. Maegok-myeon's regional industrial structure has shown a decentralized trend because the proportion of manufacturing declined from 86.74% in 2010 to 62.93% in 2015. In 2015, the raw-material recycling industry increased to 17.22%, becoming a newly emerging industry. The reason for the decentralization of the industrial structure in the Yonghwa-myeon region is the sharp increase in the healthcare industry, which rose from 2.5% in 2010 to 70.2% in 2015.

Through the above analysis, it can be determined that the primary reason for the change in industrial structure in rural areas is the evolving scale of the original major industries, whereas the major industry in most areas is manufacturing. With the expansion of the scale of manufacturing, the regional industrial structure is more concentrated. With a reduction in the manufacturing scale, the regional industrial structure is decentralized. The changing trends of the industrial structure in the rural areas of Chungbuk can be roughly divided into two categories. First, the scale of the manufacturing industry has expanded, while that of other industries has shrunk; that is, the region has further developed its focus on the manufacturing industry. Second, the scale of the manufacturing industry has shrunk sharply, while that of the service industry has expanded; that is, the region has developed many dispersed industries.

Table 5 summarizes the regions in which the industrial structure has changed. The industrial structure is concentrated in the following 11 regions: Maepo-eup, Bongyang-eup, Cheongpung-myeon, Susan-myeon, Daesowon-myeon, Suanbo-myeon, Wonnam-myeon, Naesu-eup, Samseung-myeon, Tanbu-myeon, and Hwanggan-myeon. There are 18 regions with decentralized industrial structures, as follows: Eosangcheon-myeon, Yeongchun-myeon, Sotae-myeon, Jungangtap-myeon, Maengdong-myeon, Chopyeong-myeon, Yeonpung-myeon, Mungwang-myeon, Sosu-myeon, Munui-myeon, Songnisan-myeon, Hoein-myeon, Gunseo-myeon, Gunbuk-myeon, Cheongseong-myeon, Cheongsan-myeon, Yonghwa-myeon, and Maegok-myeon.

Table 5. Summary table of regions with changes in industrial structure.

Industrial Structure Change Trend	Regions
Regions with concentrated industrial structure	Maepo-eup, Bongyang-eup, Cheongpung-myeon, Susan-myeon, Daesowon-myeon, Suanbo-myeon, Wonnam-myeon, Naesu-eup, Samseung-myeon, Tanbum-yeon, and Hwanggan-myeon.
Regions with decentralized industrial structure	Eosangcheon-myeon, Yeongchun-myeon, Sotae-myeon, Jungangtap-myeon, Maengdong-myeon, Chopyeong-myeon, Yeonpung-myeon, Mungwang-myeon, Sosu-myeon, Munui-myeon, Songnisan-myeon, Hoein-myeon, Gunseo-myeon, Gunbuk-myeon, Cheongseong-myeon, Cheongsan-myeon, Yonghwa-myeon, and Maegok-myeon.

3.3. Analysis of the Relationship between Industrial Structure and Economic Development in Rural Areas

3.3.1. Two-Step Cluster Analysis

Generally, the main causality analysis method is regression analysis, but there are limited studies on the relationship between industrial structure and economic development in rural areas. Therefore, this study is an exploratory analysis. The possible relationship between industrial structure and regional economic development in rural areas is analyzed by indirect analysis; that is, the possible relationship between them is indirectly reflected by analyzing the changes in the cluster results of regions where industrial structure has changed. In this study, the sample data of 102 rural areas in Chungcheongbuk-do were entered into SPSS for a two-step cluster analysis. The number of clusters was automatically determined using the BIC value (Schwartz's Bayesian inference criterion) as the cluster criterion. Noise processing was implemented to eliminate outliers, and log-likelihood distance was used to calculate the distance. Finally, according to the BIC, the analysis results were divided into two clusters. The results are shown in Tables 6 and 7. GIS was used to draw a distribution map of the cluster results for 2010 and 2015, as shown in Figures 4 and 5.

Table 6. Cluster results of 2010.

	Cluster 1 (<i>n</i> = 48, 47.1%)	Cluster 2 (<i>n</i> = 54, 52.9%)
Definition of the cluster	Concentrated high employment	Decentralized low employment
Second industry	Wholesale and retail (97.9%)	Education (16.7%)
Third industry	Educational (20.8%)	Wholesale and retail (35.2%)
HHI	Mean 0.57	Mean 0.36
First industry	Manufacturing (93.8%)	Manufacturing (59.3%)
Number of employees	Mean 3019	Mean 1729

Table 7. Cluster results of 2015.

	Cluster 1 (<i>n</i> = 49, 48%)	Cluster 2 (<i>n</i> = 53, 52%)
Definition of the cluster	Concentrated high employment	Decentralized low employment
Second industry	Wholesale and retail (98%)	Manufacturing (18.9%)
Third industry	Educational (16.3%)	Wholesale and retail (41.5%)
First industry	Manufacturing (93.9%)	Manufacturing (43.4%)
HHI	Mean 0.51	Mean 0.36
Number of employees	Mean 4567	Mean 1817

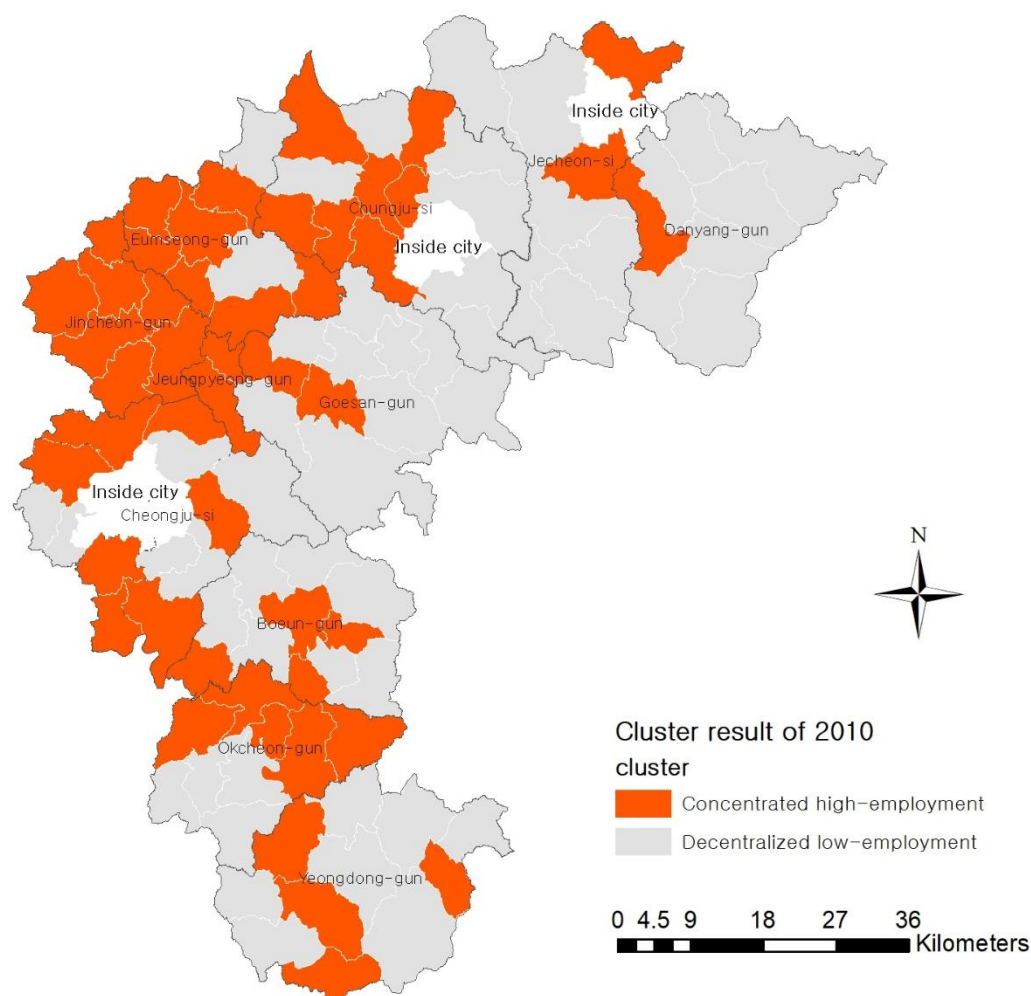


Figure 4. Distribution of clustering results of 2010.

For the 2010 data, the system automatically categorized the 102 regions into two clusters. There were 48 regions in the first cluster, accounting for 47.1%. In this cluster, the regions with manufacturing as the primary industry account for 93.8%, that is, most of the regions' primary industry is manufacturing; the regions with wholesale and retail industries account for 97.9%, that is, the majority of the second industry is wholesale and retail. There are many types of third industries that are scattered, but the number of regions with the educational service industry as the third industry is the largest, accounting for 20.8%. Therefore, in the first cluster, the main industries in the regions were manufacturing, wholesale, and retail. The average HHI is 0.57, that is, the industrial structure is relatively concentrated. The average number of employees is 3019.

There were 54 regions in the second cluster, accounting for 52.9%. In this cluster, the types of first industries are more dispersed, and are not concentrated in a certain industry, but the number of regions where manufacturing is the first industry is the largest, accounting for 59.3%. The clusters of second industries are more decentralized, but the number of regions with education services as the second industry is the largest, accounting for 16.7%. The types of third industries are also relatively diversified, but the number of regions with wholesale and retail industries is the largest, accounting for 35.2%. Therefore, compared with the first cluster, in the second cluster the top three industries in each region are not similar. The average HHI is 0.36, meaning that the industrial structure is relatively decentralized. The average number of employees is 1729.

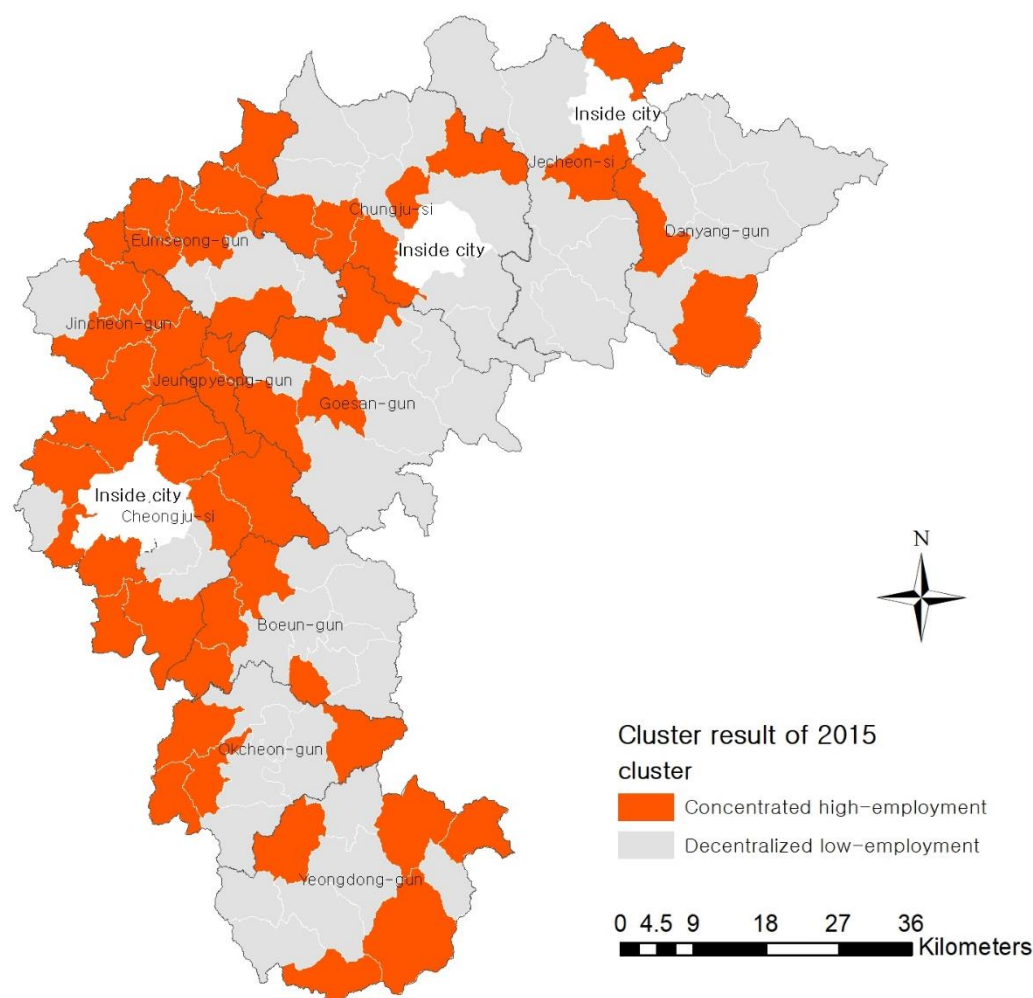


Figure 5. Distribution of clustering results of 2015.

Through comparative analysis, it was found that the average HHI of the regions in the first cluster was 0.57, and the industrial structure was relatively concentrated, while the average HHI of the regions in the second cluster was 0.36, and the industrial structure was relatively decentralized. The average number of employees in the first cluster is 3019, while the average number of employees in the second cluster is 1729. The average number of employees in the first cluster is much higher than that in the second cluster. Therefore, the first cluster can be characterized as the concentrated high employment cluster, and the second can be characterized as the decentralized low employment cluster.

For the 2015 data, the system automatically divided the 102 regions into two clusters. The first cluster contained 49 regions, accounting for 48% of the total. In this sub-cluster, the regions with manufacturing as the first industry account for 93.9%, that is, most of the regions' first industry is manufacturing. The regions with wholesale and retail industries as the second industry account for 98%, that is, most of the second industries are wholesale and retail. There are many types of third industries that are scattered, but the number of regions with the educational service industry as the third industry is the largest, accounting for 16.3%. Therefore, in the first cluster, the main industries in the regions were manufacturing, wholesale, and retail. The average HHI is 0.51, meaning that the industrial structure is relatively concentrated. The average number of employees is 4567.

The second cluster contained 53 regions, accounting for 52% of the total. In this cluster, the types of first industries are relatively decentralized and do not focus on a specific industry. However, the number of regions with manufacturing as the first industry is the largest, accounting for 43.4%. The types of secondary industry are more decentralized,

as are the types of third industry. However, the number of regions with wholesale and retail as third industries is the largest, accounting for 41.5%. Therefore, compared to the first cluster, the top three industries in each region are different. The average HHI is 0.36, meaning that the industrial structure is relatively decentralized. The average number of employees is 1817.

Through a comparative analysis, it can be found that the average HHI of the regions in the first cluster is 0.52, which suggests that the industrial structure is relatively concentrated, and the average HHI of the regions in the second cluster is 0.36, implying that the industrial structure is relatively decentralized. The average number of employees in the first cluster is 4478, while it is 1849 in the second cluster; that is, the average number of employees in the first cluster is twice as much as that in the second cluster. Therefore, the first cluster is named the concentrated high employment cluster, and the second cluster the decentralized low employment cluster.

3.3.2. Comparative Analysis of Cluster Results in Regions Where the Industrial Structure Has Changed

To analyze the relationship between industrial structure and regional economic development in rural areas, based on the above cluster results, this study analyzes the changes in cluster results in areas where the industrial structure has changed, and explains the possible relationship between them through changes in cluster results. There are 29 regions where the industrial structure has changed, and 11 regions where the industrial structure has been concentrated, as follows: Maepo-eup, Bongyang-eup, Cheongpung-myeon, Susan-myeon, Daesowon-myeon, Suanbo-myeon, Wonnam-myeon, Naesu-eup, Samseung-myeon, Tanbu-myeon, and Hwanggan-myeon. The industrial structure has been decentralized in the following 19 regions: Eosangcheon-myeon, Yeongchun-myeon, Sotae-myeon, Jungangtap-myeon, Maengdong-myeon, Deoksan-myeon, Chopyeong-myeon, Yeonpung-myeon, Mungwang-myeon, Sosu-myeon, Munui-myeon, Songnisan-myeon, Hoein-myeon, Gunseo-myeon, Gunbuk-myeon, Cheongseong-myeon, Cheongsan-myeon, Yonghwa-myeon, and Maegok-myeon.

Among the 11 regions where the industrial structure has been concentrated, although the cluster results of most regions have not changed, the Naesu-eup and Hwanggan-myeon regions registered a change, from the second cluster in 2010 to the first cluster in 2015, that is, from a decentralized low employment cluster to a concentrated high employment cluster. When the economy develops well, employment also increases. The transformation from low to high employment demonstrates that the economic development trend is satisfactory, indicating that the concentrated industrial structure has the potential to promote economic development in rural areas. The changes in the cluster results for the regions where the industrial structure has been concentrated are shown in Table 8. Among the 18 regions in which the industrial structure has been decentralized, the cluster results of 8 regions have changed. First, the cluster results of Jungangtap-myeon, Maengdong-myeon, Cheongseong-myeon, Yonghwa-myeon, and Maegok-myeon changed from the first cluster in 2010 to the second in 2015, that is, from a concentrated high employment cluster to a decentralized low employment cluster. When economic development declines, unemployment increases, meaning that the number of employees decreases. The transformation from high to low employment shows an economic recession to a certain extent, indicating that the decentralized industrial structure in rural areas has the potential to hinder economic development. Second, the cluster results for the Sosu-myeon, Hoein-myeon, and Gunseo-myeon regions changed from the second cluster in 2010 to the first cluster in 2015, that is, from the decentralized low employment cluster to the concentrated high employment cluster. The explanation for this may be that, even in rural areas, it is not ruled out that, because of their own advantages, the decentralized industrial structure is more suitable for the needs of local economic development. The changes in the cluster results for regions where the industrial structure has been decentralized are shown in Table 9.

Table 8. Regions where the industrial structure has been concentrated: changes of the cluster result.

Regions with Concentrated Industrial Structure	Cluster Results of 2010	Cluster Results of 2015
Maepo-eup	cluster 2	cluster 2
Bongyang-eup	cluster 2	cluster 2
Cheongpung-myeon	cluster 2	cluster 2
Susan-myeon	cluster 2	cluster 2
Daesowon-myeon	cluster 1	cluster 1
Suanbo-Myeon	cluster 2	cluster 2
Wonnam-myeon	cluster 1	cluster 1
Naesu-eup	cluster 2	cluster 1
Samseung-myeon	cluster 1	cluster 1
Tanbu-myeon	cluster 2	cluster 2
Hwanggan-myeon	cluster 2	cluster 1

Table 9. Regions where the industrial structure has been decentralized: changes of the cluster result.

Regions with Decentralized Industrial Structure	Cluster Results of 2010	Cluster Results of 2015
Eosangcheon-myeon	cluster 2	cluster 2
Yeongchun	cluster 2	cluster 2
Sotae-myeon	cluster 2	cluster 2
Jungangtap-myeon	cluster 1	cluster 2
Maengdong-myeon	cluster 1	cluster 2
Chopyeong-myeon	cluster 1	cluster 1
Yeonpung-myeon	cluster 2	cluster 2
Mungwang-myeon	cluster 1	cluster 1
Sosu-myeon	cluster 2	cluster 1
Munui-myeon	cluster 1	cluster 1
Songnisan-myeon	cluster 2	cluster 2
Hoein-myeon	cluster 2	cluster 1
Gunseo-myeon	cluster 2	cluster 1
Gunbuk-myeon	cluster 1	cluster 1
Cheongseong-myeon	cluster 1	cluster 2
Cheongsan-myeon	cluster 1	cluster 1
Yonghwa-myeon	cluster 1	cluster 2
Maegok-myeon	cluster 1	cluster 2

4. Conclusions

The purpose of this study is to analyze the changing trend of the industrial structure in rural areas and the possible relationship between industrial structure and economic development. Accordingly, this study considers 102 rural areas in Chungcheongbuk-do as the research object, extracts the industrial sales and the number of employees in the rural areas of Chungbuk in 2010 and 2015 according to the general economic survey provided by the National Bureau of Statistics of Korea, and conducts an empirical analysis.

First, according to the diversified industrial agglomeration theory of Jacobs (1969), based on industrial sales data in rural areas, the Herfindahl–Hirschman Index method is used to calculate the HHI value of each region, and the changing trend of regional industrial structure is analyzed by comparing the HHI value. The analysis results show that compared with 2010, the industrial structure of 73 rural areas (accounting for 71.6%) did not change, and the industrial structure of only 29 areas (accounting for 28.4%) changed in 2015. Among them, there are 11 regions with a concentrated industrial structure, as follows: Maepo-eup, Bongyang-eup, Cheongpung-myeon, Susan-myeon, Daesowon-myeon, Suanbo-Myeon, Wonnam-myeon, Naesu-eup, Samseung-myeon, Tanbu-myeon, and Hwanggan-myeon, and 18 regions with decentralized industrial structures, as follows: Eosangcheon-myeon, Yeongchun-myeon, Sotae-myeon, Jungangtap-myeon, Maengdong-myeon, Chopyeong-myeon, Yeonpung-myeon, Mungwang-myeon, Sosu-myeon, Munui-myeon, Songnisan-myeon, Hoein-myeon, Gunseo-myeon, Gunbuk-myeon, Cheongseong-myeon, Cheongsan-

myeon, Yonghwa-myeon, and Maegok-myeon. The main reason for the change in the industrial structure in rural areas is the substantial increase or decrease in the scale of the original main industries. Manufacturing is the primary industry in most areas. The scale of the manufacturing industry has expanded, the regional industrial structure is more concentrated, the scale of the manufacturing industry has shrunk, and the industrial structure of the region has become decentralized.

Second, to further explore the relationship between industrial structure and economic development in rural areas, the possible relationship between them is indirectly reflected by analyzing the changes in cluster results in areas where industrial structure has changed. Taking the top three industries as categorical variables and the HHI value and number of employees as continuous variables, a two-step cluster analysis was carried out, and we compared the cluster results in areas with changes in industrial structure. The analysis results show that, first, among the 11 regions where the industrial structure is concentrated, the cluster results of the Naesu-eup and Hwanggan-myeon regions changed from a decentralized low employment cluster in 2010 to a concentrated high employment cluster in 2015, while the cluster results in other regions remain unchanged. Second, among the 18 regions with a decentralized industrial structure, the cluster results of Jungangtap-myeon, Maengdong-myeon, Cheongseong-myeon, Yonghwa-myeon, and Maegok-myeon changed from a concentrated high employment cluster in 2010 to a decentralized low employment cluster in 2015, while the industrial structure of Sosu-myeon, Hoein-myeon, and Gunseo-myeon regions changed from a decentralized low employment cluster in 2010 to a concentrated high employment cluster in 2015. The industrial structure of other regions has not changed. Therefore, it can be concluded that, for the general rural areas, compared with the industrial structure with a high industrial diversification level, the low industrial diversification level, that is, the concentrated industrial structure, is more helpful in promoting the economic development of rural areas. However, in some special cases, for example in rural areas with certain specific advantages, the industrial structure with a high diversification level, that is, the decentralized industrial structure, is more conducive to the development of the regional economy. The research results are similar to the results of the impact of industrial diversification on the economic development of small cities, which can further support that the research results of this paper are robust.

Therefore, to promote the economic development of rural areas, the Korean and local governments should first identify the specific conditions of rural areas when formulating industrial development policies in these areas. For general rural areas, they should focus on the concentrated industrial structure and consider the actual conditions of each rural area to focus on the development of one or several industries, so as to promote the development of the regional economy. For special rural areas, they should combine the advantages of the region and adopt the decentralized industrial structure development strategy to meet the development needs of industrial diversification in special rural areas. Specifically, when the Korean government and others formulate industrial development strategies in rural areas, they should adapt to local conditions and formulate differentiated industrial development strategies rather than generalized, unified industrial development strategies. However, this paper also has the following limitation, which is that it does not clarify what kind of areas “special rural areas” are. There is no currently no standard with which to distinguish between “general rural areas” and “special rural areas”. Therefore, the follow-up research should focus on establishing such a standard to distinguish between “general rural areas” and “special rural areas”. After screening the nature of rural areas, more targeted industrial development strategies can be put forward.

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